SHERPA: Joint Estimation of Used Heavy-Equipment Sale Prices and Rental Rates from Sparse Marketplace Listings

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Abstract

Online marketplaces for construction and industrial machinery publish asking prices for sale and asking rates for hire, but their listings are few per category, often incomplete, and noisy. We present SHERPA (Shrinkage-based Heavy Equipment Residual Price Analysis), an explainable model that fits sale prices and rental rates *jointly*: both are treated as observations of one underlying machine value, built from a flexible Box–Cox depreciation curve, size, usage relative to age, condition, attachments, seller type, a brand-to-origin hierarchy and further listing attributes. The rental side adds a per-category rent-to-value level, motivated by the user cost of capital, a duration discount under which the rate per day falls as a power of hire length, and an operator premium. Every effect is estimated robustly (medians and Theil–Sen slopes) by backfitting with empirical-Bayes partial pooling and domain constraints, stabilised by averaging the iterates, and every estimate carries a cross-conformal prediction interval calibrated separately for listings with missing inputs. In a simulation study on synthetic markets with known parameters, SHERPA recovers the parameters it was given and is the most accurate model at every market size: with 20 listings per category and side it reduces the median absolute percentage error of sale-price estimates from 45% for an ordinary least-squares hedonic regression to 31%, and of rental rates from 50% to 28%. It is insensitive to impossible prices that more than double the regression's error, fitting rental listings jointly cuts sale-price error by up to 11 percentage points when sales are scarce, and its conformal intervals stay conservative in small markets where normal intervals under-cover. The model is released as open-source software in TypeScript and Python.

1 Introduction

What is a ten-year-old 20-tonne excavator worth, and what should it rent for by the day, the week or the month? Contractors, rental companies, lenders and insurers ask these questions daily. The academic literature on the residual value of heavy equipment answers the first from auction records with clean, structured fields (Lucko et al., 2006; Fan et al., 2008; Milošević et al., 2021); the second is usually answered by rules of thumb from trade rate guides. Neither setting matches the data most regional equipment markets actually produce: listings on an online marketplace, where a category may have a few dozen priced machines, the manufacture year sits in the title if anywhere, engine hours are rarely given, and a placeholder price of 1 or a monthly rate typed into the sale field is common.

This paper presents SHERPA, a model built for that data. Its premise is that a sale listing and a rental listing of the same kind of machine are two observations of one underlying value: the rental rate is the price of using the machine for a while, and capital theory links it to the value of the machine through the user cost of capital (Jorgenson, 1963; Hall and Jorgenson, 1967). Fitting both

together lets a market with many rental listings and few sales still learn how age, size, condition or brand move value, and the reverse.

Contributions.

1. A hedonic model with *one machine-value core identified by two equations*, one for sale prices and one for rental rates, with a per-category rent-to-value level, a power-law duration discount and an operator premium on the rental side (Section 3).

2. An estimation procedure for sparse, messy listings: robust backfitting with partial pooling over a category-family and brand-origin hierarchy, sign and monotonicity constraints, an identification strategy that prevents the sale and rent levels from drifting together, and averaging of the iterates for a stable answer (Section 4).

3. Cross-conformal prediction intervals calibrated separately by side and by the number of missing inputs (Section 4.1).

4. A simulation study with known parameters: recovery, accuracy against ordinary least squares and a simpler baseline, robustness to impossible prices and missing fields, the value of rental listings for sale estimates, interval calibration and ablations (Section 6).

5. Open-source TypeScript and Python implementations without dependencies, kept equal by a cross-language test (Section 8).

All experiments use synthetic markets, so that every estimate can be compared with the truth. A case study on real marketplace listings is left to a later version of this paper.

2 Related work

Residual value of machinery. Statistical models of the residual value of heavy construction equipment go back to Lucko et al. (2006) and Lucko et al. (2007), who regressed auction prices on age, hours and economic conditions, revisited after the 2008 recession by Lucko (2011). Fan et al. (2008) used regression trees; more recent work applies ensembles and neural networks to large sets of auction or dealer records (Shehadeh et al., 2021; Alshboul et al., 2021; Milošević et al., 2021; Toma et al., 2024), with Huymajer et al. (2025) cautioning against their pitfalls, and Chiteri (2018) relating residual value to equipment cash flows. In agriculture, remaining-value functions for tractors and implements have a long history (Cross and Perry, 1995, 1996; Dumler et al., 2003; Wu and Perry, 2004) and inform the ASABE D497 standard (ASABE, 2011); Herranz-Matey and Ruiz-Garcia (2023, 2024) revisit them for European tractors. These models price sales only, and none reports calibrated intervals.

Depreciation and functional form. Hulten and Wykoff (1981) estimated economic depreciation from vintage asset prices with Box–Cox forms (Box and Cox, 1964), letting the data choose between geometric, linear and concave patterns; Wu and Perry (2004) compared functional forms for farm machinery and Halvorsen and Pollakowski (1981) for hedonic equations generally. SHERPA adopts a Box–Cox curve in age with a shape parameter per category.

Hedonic prices and rents. Hedonic regression (Rosen, 1974) and regression price indices (Bailey et al., 1963) are the standard tools for pricing differentiated goods over time. Joint attention to prices and rents comes mostly from housing: Bracke (2015) matches London homes that were both sold and rented, Hill and Syed (2016) compare hedonic price–rent ratios with the user cost, and Crone et al. (2004) and Garner and Verbrugge (2009) relate owner and rental costs. In vehicles, Cho and Rust (2010) study how rental prices of cars depend on age and mileage. Guasch and Marshall (1987) document a length-of-residency discount in rents, the housing analogue of the hire-duration discount modelled here. We found no published model that fits sale prices and rental rates of used equipment jointly with one shared value function, nor a data-driven estimate of the duration discount for equipment; trade guides state such multipliers as rules of thumb.

Seller type. Used-car markets show systematic dealer premiums (Genesove, 1993; Emons and Sheldon, 2009; Biglaiser et al., 2020), which vary with the type of intermediary (Li et al., 2025). SHERPA includes seller type as a level effect.

Shrinkage. Pulling small-group estimates toward a parent is the empirical-Bayes idea of James and Stein (1961) and Efron and Morris (1973, 1975), used in small-area estimation (Fay and Herriot, 1979), multilevel regression (Gelman and Hill, 2007), house-price models (Brunauer et al., 2013; Gelfand et al., 2004; Goodman and Thibodeau, 2003) and target encoding of high-cardinality categories (Micci-Barreca, 2001).

Robust estimation. SHERPA relies on medians, the Theil–Sen slope (Theil, 1950; Sen, 1968), the median absolute deviation as a scale (Hampel, 1974; Rousseeuw and Croux, 1993), and the robustness ideas of Huber (1964), fitted by backfitting (Hastie and Tibshirani, 1986, 1990) with averaged iterates (Polyak and Juditsky, 1992).

Prediction intervals. Conformal prediction gives distribution-free intervals (Vovk et al., 2005; Lei et al., 2018), with cross-validation variants analysed by Barber et al. (2021) and quantile-based variants by Romano et al. (2019). Normalised nonconformity scores have been applied to automated valuation of real estate (Bellotti, 2017; Lim and Bellotti, 2021; Bastos and Paquette, 2025; Hjort et al., 2025); we are not aware of applications to equipment prices.

3 Model

3.1 Data

A listing i is either a sale, with an asking price P_i, or a rental, with an asking rate for a hire of T_i days (a daily, weekly or monthly rate). It describes a machine of category c in family $f(c)$ by any subset of: brand b and its origin group $o(b)$, manufacture year (hence age a), size s in the category's unit (tons, kVA), engine hours h, new or used, a condition grade $q \in \{1, 2, 3, 4\}$, a set of attachments A, an emissions standard e, the remaining life of wear parts $w \in [0, 1]$, a spec or import status g, a seller type k, an inspection flag v, a region r, the listing month t, months on the market m and a completeness score ℓ. Every field except the category, the side and the price may be missing. One model is fitted per market.

3.2 Machine value

The log value of machine x is a sum of named effects around a reference machine of category c (median age and size, used, good condition, no attachments):

$$V(x) = \alpha_c + \beta_c \left(g_{\lambda_c}(a) - g_{\lambda_c}(a_c^\star)\right) + \gamma_c \log \frac{s}{s_c^\star} + \delta_c\, u(h,a) + \theta_c\, \mathbf{1}[\text{new}] + \eta_q + \sum_{j \in A} \psi_j$$
$$+ \varepsilon_e + \zeta_c \left(w - \tfrac{1}{2}\right) + \varphi_g + \kappa_k + \nu\, v + b_{o(b)} + b_b + b_{b,c} + \rho_r + \tau_t. \tag{1}$$

The depreciation curve is a Box–Cox transform of age,

$$g_\lambda(a) = \frac{(1+a)^\lambda - 1}{\lambda}, \qquad g_0(a) = \log(1+a), \qquad \lambda \in [-1, 1], \tag{2}$$

which is linear in age (geometric depreciation) at $\lambda = 1$ and front-loaded for smaller λ. Usage enters relative to what is normal for the age,

$$u(h,a) = \log \frac{h + h_0}{\bar{h}_c \max(a, \tfrac{1}{2}) + h_0}, \tag{3}$$

with $\bar{h}_c$ the category's typical annual hours and $h_0 = 250$, so that hours do not duplicate the age effect. Categorical sets without a natural reference level (emissions, spec, seller, origin, region) include an explicit *unknown* level and are centred on the average listing; grade is referenced at "good" ($\eta_3 = 0$).

Domain knowledge enters as constraints: machines lose value with age ($\beta_c < 0$), larger machines cost more ($\gamma_c \geq 0$), more usage and worse condition never raise the price ($\delta_c \leq 0$, $\eta_1 \leq \eta_2 \leq \eta_3 \leq \eta_4$), and new machines and attachments never lower it ($\theta_c \geq 0$, $\psi_j \geq 0$).

3.3 Sale and rent equations

Sale prices and rental rates share V:

$$\log P_i = V(x_i) + \lambda_m \log(1 + m_i) + \lambda_\ell(\ell_i - \tfrac{1}{2}) + \epsilon_i^{\text{sale}}, \tag{4}$$

$$\log R_i(T_i) = V(x_i) + \pi_c + \kappa_c \log \frac{T_i}{30} + \omega_c\, \mathbf{1}[\text{operator}] + \lambda_m \log(1 + m_i) + \epsilon_i^{\text{rent}}, \tag{5}$$

where $R_i(T)$ is the asking rate per day for a hire of T days. The listing terms describe the listing, not the machine, and are set to their reference values when a machine is valued.

π_c is the log rate per day per unit of value at a one-month hire. Capital theory prices the use of an asset by its user cost, $c = q\,(r + \delta) - \dot{q}$ for an asset of price q, interest rate r and depreciation rate δ (Jorgenson, 1963; Hall and Jorgenson, 1967); categories that wear fast or stand idle between hires therefore rent dearer relative to their value. $\kappa_c \leq 0$ is the duration discount: the rate per day for a T-day hire is $R(T) = R(30)\,(T/30)^{\kappa_c}$, so one parameter per category turns a daily rate into weekly and monthly ones. $\omega_c \geq 0$ is the premium for a wet hire with an operator.

3.4 Hierarchy

Every group estimate is shrunk toward its parent by partial pooling,

$$\hat{\theta}_{\text{group}} = \frac{n}{n+k}\, \tilde{\theta}_{\text{group}} + \frac{k}{n+k}\, \hat{\theta}_{\text{parent}}, \tag{6}$$

where $\tilde{\theta}$ is the group's own robust estimate from n listings and k the number of listings at which group and parent weigh equally. Table 1 lists the structure.

Table 1: Hierarchy, default shrinkage strengths and constraints.

Parameter	Group	Parent	k	Constraint
α	category	family	6	–
$D = \beta g_\lambda(H)$, λ	category	pooled	12, 24	$D < 0$, $\lambda \in [-1, 1]$
γ, δ, ζ	category	pooled	12	$\gamma \geq 0$, $\delta \leq 0$, $\zeta \geq 0$
θ	category	pooled	10	$\theta \geq 0$
η_q	grade	0	8	monotone
ψ_j	attachment	0	6	$\psi \geq 0$
$\varepsilon, \varphi, \kappa_k, \rho$	level	0	8	centred
b_o, b_b, $b_{b,c}$	origin, brand, brand×category	0	6, 4, 6	centred (b_o)
π, κ, ω	category	pooled	8	$\kappa \leq 0$, $\omega \geq 0$
σ	category×side	pooled	8	$\sigma \geq 0.1$

Algorithm 1 Fitting SHERPA

Drop non-positive prices and log prices more than 3.5 robust standard deviations from the median of their category and side (rates put on a 30-day footing first).
Initialise α_c at the median sale log price, π_c at the rent–sale median gap.
for round $= 1, \ldots, 40$ **do**
    for each block B in turn **do**
        $\tilde{y}_i \leftarrow y_i - \sum_{B' \neq B} \text{term}_{B'}(x_i)$ ▷ partial residuals
        estimate B by medians or Theil–Sen slopes of $\tilde{y}$; shrink; clamp
    end for
    centre the reference-free level sets and the time index; move each shift into α_c
    if no prediction moved by more than 10^{-4} **then** stop
    if round ≥ 15 **then** add all parameters to a running average
end for
Set the parameters to their average over the rounds after the burn-in.
$\sigma_{c,\text{side}} \leftarrow 1.4826\times$ MAD of residuals, shrunk to the pooled scale.
Calibrate intervals by cross-conformal scores (Section 4.1).

4 Estimation

SHERPA is fitted by robust backfitting (Algorithm 1): each block of parameters is re-estimated on the partial residuals of all the others, using only medians and Theil–Sen slopes, then shrunk and clamped.

Identification of the two levels. α_c and π_c are separately identified only where a category has both sale and rental listings. We estimate α_c from sale listings alone wherever a category has at least three, and π_c from rental listings given α_c; elsewhere π_c takes its pooled value. Estimating α_c from both sides let α_c and π_c slide along a ridge: the shrinkage of π_c toward the pooled value pulled it down a little each round, α_c rose to compensate through the rental listings, and sale listings ended up over-predicted.

The depreciation curve. β_c changes its units with λ_c, so the shrunk quantity is the depreciation at a fixed horizon, $D_c = \beta_c\, g_{\lambda_c}(H)$ with $H = 10$ years. λ_c is chosen by profiling the L_1 fit of a Theil–Sen line over a grid in $[-1, 1]$, refined by a parabola through the best grid point and its neighbours, during the first five rounds only, and then held fixed.

Averaging. Backfitting with median updates is not guaranteed to converge. In our experiments it reached the neighbourhood of a fixed point and then wandered around it by about ±0.8% in prediction indefinitely; its answer depended on the round at which it stopped. Averaging the iterates after a burn-in, as in stochastic approximation (Polyak and Juditsky, 1992), fixes the answer and reduces the round-to-round movement about tenfold.

4.1 Prediction intervals

For an interval at level $1-\alpha$, the listings are split into K folds; each fold is predicted by a model fitted on the others, and its normalised scores $S_i = |\log y_i - \log \hat{y}_i| / \hat{\sigma}_{c,\text{side}}$ are pooled by side and by the number of missing price-moving inputs (year, size, hours, grade: 0, 1, or 2 or more), a Mondrian split. The interval is $\hat{y}\exp(\pm\hat{q}\,\hat{\sigma})$ with $\hat{q}$ the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score of the group (Vovk et al., 2005; Lei et al., 2018). Listings with gaps are thus calibrated against listings with gaps. Cross-conformal intervals carry a slightly weaker worst-case guarantee than split-conformal ones (Barber et al., 2021); we report their empirical coverage.

5 Simulation design

Synthetic markets. Each market has five categories in three families (excavators, wheel loaders and compactors in earthmoving, cranes in lifting, generators in power), eight brands in five origin groups, and sale and rental listings drawn from Equations (1)–(5) with known parameters (Appendix A). Ages are uniform on 1–15 years with 12% new machines on the sale side; hours are log-normal around the category's typical use; rental periods are 1, 7 or 30 days, 30% with an operator. The markets are then made as messy as marketplace data: by default 20% of years, sizes, grades, specs, sellers and regions are left empty, half of all hours and 60% of emissions and wear fields are empty, and 2% of prices are wrong by a factor of 1,000. Noise has a standard deviation of 0.22–0.30 on log price. Every brand, region and number is invented.

Baselines. (i) The median of the training listings of the same category and side (and hire period). (ii) SHERPA-0, a simpler sale-only model: per-category log price with Theil–Sen age and size slopes, a new-machine premium and brand effects, shrunk to pooled slopes. (iii) An ordinary least-squares hedonic regression on the same inputs: category intercepts, category-specific age and log-size slopes, rent-side intercepts per category, log hire period and operator, dummy variables for every level, indicators for missing inputs, and a small ridge penalty.

Protocol. Four-fold cross-validation, repeated over independently generated markets. Models are fitted on the raw training listings, junk prices included; they are scored on held-out listings that survive the cleaning step, since an impossible price is not one any model should be judged on. The metric is the median absolute percentage error (MdAPE) of the central estimate; intervals are scored by the share of held-out prices they contain.

6 Results

6.1 Parameter recovery

Table 2 reports the bias and root-mean-square error of every parameter over 20 synthetic markets with 120 sale and 120 rental listings per category.

Table 2: Parameter recovery over 20 synthetic markets (120 sale and 120 rental listings per category). Errors are on the log-price scale; level effects without a reference level are compared after centring. n counts estimates (parameters × markets).

Parameter	Bias	RMSE	n
depreciation (D)	0.006	0.207	100
curve shape (λ)	0.010	0.321	100
size elasticity (γ)	−0.000	0.048	80
usage (δ)	0.013	0.069	100
new premium (θ)	0.140	0.203	100
wear (ζ)	0.025	0.150	100
rent-to-value (π)	−0.042	0.112	100
duration discount (κ)	−0.002	0.027	100
operator premium (ω)	0.005	0.111	100
condition grade (η)	0.016	0.051	60
attachments (ψ)	0.002	0.025	80
seller type (κ_k)	−0.000	0.030	60
brand origin	−0.000	0.036	100
emissions (ε)	−0.000	0.027	80
spec (φ)	0.000	0.014	40
region (ρ)	−0.000	0.022	60
inspection (ν)	0.002	0.021	20
time on market (λ_m)	−0.017	0.037	20
completeness (λ_ℓ)	0.002	0.051	20
market trend / month (τ)	−0.001	0.003	20

Nearly every parameter is recovered without material bias. The duration discount κ (RMSE 0.027), size elasticity (0.048), attachment premiums (0.025), the level effects of seller type, brand origin, emissions, spec and region (0.014–0.036 after centring) and the monthly market trend (0.003) are recovered closely; the rent-to-value level π has a small negative bias (−0.04) inherited from the shrinkage of a parameter identified only through the difference of two levels. Two parameters behave as Section 7 anticipates: the curve shape λ is unbiased on average but noisy (RMSE 0.32), and the new-machine premium θ is biased upward (+0.14), because new machines are observed only at age 0.

Figure 1 shows the fitted curves of one market against the truth. The duration discount is recovered almost exactly; the depreciation curves follow the true ones within the spread of the data, with the noisier shape estimates visible in the first years.

6.2 Accuracy and market size

Table 3 compares the models as the number of listings per category and side grows.

SHERPA-1 has the lowest error at every market size, on both sides. The gap is largest in small markets: with 20 listings per category and side, ordinary least squares is no better than the category median (45.4% against 46.2% on sale prices), while SHERPA-1 reaches 30.6%; on rental rates the difference is 49.6% against 28.0%. With 160 listings the hedonic regression has enough data for its many dummy variables and closes most of the gap (29.4% against 24.2% on sale prices). SHERPA-0 is competitive with ordinary least squares on sale prices but cannot price rentals.

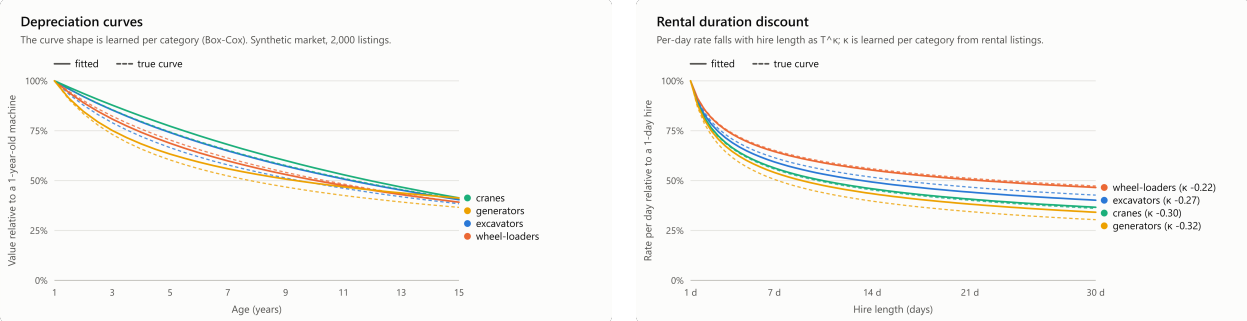


Figure 1: Fitted (solid) and true (dashed) depreciation curves (left) and rental duration discounts (right) for four categories of one synthetic market of 2,000 listings.

Table 3: Median absolute percentage error (%) by listings per category and side; four-fold cross-validation, mean over three markets. Best per side in bold.

	Sale prices				Rental rates		
n	Median	SHERPA-0	OLS	SHERPA-1	Median	OLS	SHERPA-1
20	46.2	33.6	45.4	**30.6**	50.5	49.6	**28.0**
40	53.1	35.9	38.1	**31.4**	50.8	44.3	**28.4**
80	50.9	31.4	33.2	**26.5**	48.0	35.3	**26.1**
160	50.2	30.0	29.4	**24.2**	45.5	30.0	**23.9**

6.3 Robustness

Table 4 varies the share of impossible prices and the share of empty fields.

On clean synthetic data the regression and SHERPA-1 are level (26.3% and 27.2%, within one standard error). A share of impossible prices—placeholders, a missing or extra factor of 1,000—is where they part: 5% of such prices raise the regression's error to 42%, 20% to 65%, worse than the category median, while SHERPA-1 stays at 26–28% because its cleaning step and median-based estimates ignore them. As fields are left empty, every model degrades, but SHERPA-1 keeps its lead (36.9% against 41.6% on sale prices with half of all fields empty).

6.4 Rental listings help sale estimates

Table 5 holds 160 rental listings per category and varies the sale listings.

Rental listings improve sale estimates, and most where sales are scarce. With 10 sale listings per category, fitting the 160 rental listings jointly lowers the sale-price error from 42.7% to 31.9%; with 20 from 37.3% to 31.5%; with 40 from 30.9% to 27.0%. The rental listings carry information about the depreciation curve, size, condition and brand effects that the shared value core passes to the sale side.

6.5 Interval calibration

Table 6 compares cross-conformal intervals with normal intervals of the same scale. In small markets (40 listings per category and side) normal intervals under-cover—44% and 73% of held-out sale prices fall inside the nominal 50% and 80% intervals—because the robust scale is estimated from few residuals and missing inputs add error the scale does not see. The conformal intervals

Table 4: Robustness: MdAPE (%) with 80 listings per category and side, mean over three markets. Models are trained on the raw listings; scored on held-out listings with plausible prices.

	Sale prices			Rental rates		
Condition	Median	OLS	SHERPA-1	Median	OLS	SHERPA-1
0% impossible prices	50.8	**26.3**	27.2	46.6	28.8	**27.4**
5% impossible prices	51.0	42.0	**27.0**	47.8	42.4	**27.6**
10% impossible prices	50.8	51.0	**26.2**	48.9	51.8	**27.6**
20% impossible prices	51.4	64.6	**27.8**	48.1	66.7	**27.9**
10% empty fields	48.8	31.9	**24.3**	48.4	32.4	**24.5**
30% empty fields	48.8	35.9	**31.3**	48.4	36.3	**29.5**
50% empty fields	48.8	41.6	**36.9**	48.4	38.3	**33.2**

Table 5: Sale-price MdAPE (%) with 160 rental listings per category and few sale listings; mean over five markets. *Sale only* fits SHERPA-1 without the rental listings.

Sale listings per category	Median	SHERPA-1, sale only	SHERPA-1, joint	Gain (points)
10	54.5	42.7	**31.9**	10.8
20	54.8	37.3	**31.5**	5.8
40	51.1	30.9	**27.0**	3.8

are conservative instead (57% and 87%). With 160 listings both are close to nominal (52–54% and 82–84%). Conformal calibration thus matters most where data are thin, which is the setting SHERPA is designed for.

6.6 Ablations

Figure 2 switches off one block at a time on a market of 1,200 listings.

The flexible depreciation curve contributes most (+5.9 points of MdAPE when replaced by a fixed curve), followed by partial pooling (+2.0), condition grade (+1.9), the brand hierarchy (+1.4), seller type (+1.2) and the joint fit of rental listings (+1.1). Attachment effects in this market are small (3–10% of value on a quarter of listings) and estimating them costs slightly more than it gains (−0.7).

7 Discussion

What the joint model buys. Sharing the value core lets each side of the market inform the other. The gain is largest where it matters most in practice: categories that are often hired and rarely sold.

Weakly identified parameters. New machines are listed at age 0 and used machines from about one year, so listings cannot separate the premium of a new machine from the curvature of the depreciation curve in its first year. In the simulations λ is unbiased on average but noisy (RMSE 0.32) and θ is biased upward (+0.14), while the ten-year depreciation D is recovered without bias. The two parameters should be interpreted together; the estimated value of a new machine is a safer summary than θ alone.

Table 6: Share (%) of held-out prices inside the 50% and 80% intervals; four-fold cross-validation, mean over three markets. Nominal levels: 50 and 80.

		Sale		Rent	
n	Intervals	50%	80%	50%	80%
40	conformal	57.1	86.6	57.1	86.4
40	normal	44.3	73.2	42.5	69.2
160	conformal	52.1	82.9	52.6	83.0
160	normal	54.3	83.5	53.7	81.7

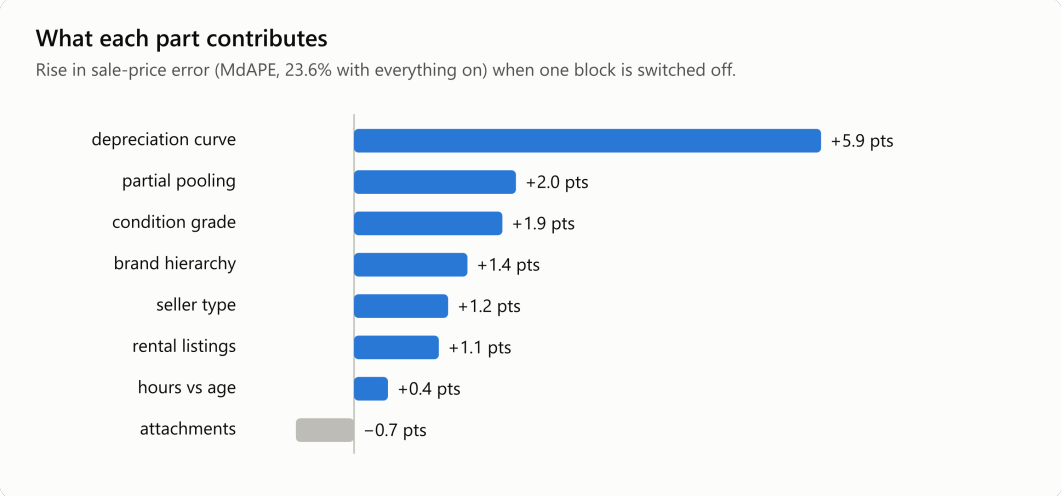


Figure 2: Rise in sale-price MdAPE when one block of SHERPA is switched off (five-fold cross-validation, 1,200 synthetic listings).

Numerical stability. Medians are step functions, so the fit responds to tiny perturbations by small jumps. With averaged iterates, a relative change of 10^{-9} to every price moves predictions by a median of 0.3%; the TypeScript and Python implementations agree to the last digit for three rounds and within the same margin thereafter.

Limitations. SHERPA models asking prices and rates, not transaction prices; the gap requires transaction data (cf. Han and Strange, 2016). Listings without a price are not missing at random. The conformal guarantee assumes exchangeability, which strong market trends weaken. All results here are on synthetic data drawn from the model's own form, which favours the model; the comparison with ordinary least squares measures the value of robustness, pooling and joint fitting, not of the functional form. A case study on real listings, and comparisons with gradient-boosted trees, are planned for the next version.

8 Software and reproducibility

SHERPA is available under the Apache License 2.0 at `https://github.com/dev-kakosoai/sherpa-valuation`. Release v0.1.0 contains a TypeScript package (`@equipmentsfinder/sherpa`) and a Python package (`sherpa-valuation`). Neither has runtime dependencies; listings and fitted models are plain JSON in both, and a cross-language test keeps the implementations equal. Every

number in this paper is produced by the scripts `bench/paper.ts` and `bench/benchmark.ts` in that repository.

Acknowledgements

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A Parameters of the synthetic markets

Table 7: Category parameters of the synthetic markets. D is the log-value change from new to age 10; π is given as the daily rate per unit of value, e^{π}.

Category	Base price	D	λ	γ	δ	θ	ζ	e^{π}	κ	ω
excavators	300,000	−0.90	0.40	0.70	−0.15	0.20	0.30	0.0013	−0.25	0.35
wheel-loaders	260,000	−0.80	0.60	0.60	−0.12	0.18	0.25	0.0012	−0.22	0.30
cranes	900,000	−0.70	0.80	0.80	−0.10	0.15	0.15	0.0011	−0.30	0.25
generators	60,000	−1.10	0.10	0.55	−0.20	0.22	0.00	0.0025	−0.35	0.00
compactors	45,000	−0.85	0.50	0.00	−0.10	0.15	0.20	0.0018	−0.28	0.20

Grade: 1: −0.45, 2: −0.18, 3: 0.00, 4: 0.12. Attachments: bucket 0.03, breaker 0.10, coupler 0.04, forks 0.05. Seller: dealer 0.10, rental −0.04, private −0.06. Origin: japan 0.08, europe 0.06, usa 0.05, korea −0.03, china −0.20. Emissions: tier2 −0.10, tier3 0.00, tier4i 0.05, tier4f 0.10. Inspection 0.06; market trend 0.004 per month; time on market 0.04; completeness 0.08. Noise standard deviation 0.22–0.30 on log price, by category.